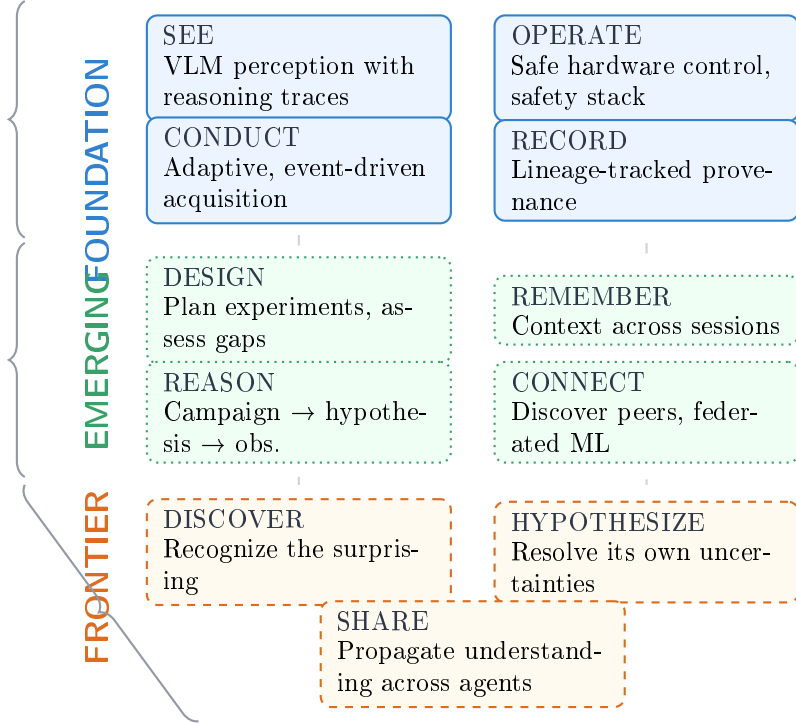




the human: operator → collaborator → supervisor

Single Instrument



perceive → reason → act

One agent governs
one microscope.
Closes the semantic gap
from pixels to biology.

Semantic gap

Multi-Instrument



Same-domain mesh.
Share data, federated ML,
cross-validate across
instruments and labs.

Coverage gap

Planetary Science



Cross-domain agents.
Microscopy, oceanography,
astronomy, genomics —
inter-agent communication.

Knowledge gap

Same architectural pattern, dynamically expanded context, empirically grown connections